Biomechanical Literature Knowledge Graph Construction via Zero-Shot NLP

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Abstract

This project constructs a biomedical knowledge graph from a manually curated dataset of 19 biomechanical study abstracts using a fully zero-shot NLP pipeline. The pipeline covers two core stages: Named Entity Recognition (NER) and Relation Extraction (RE). For NER, a two-round extraction approach is applied using the pre-trained model GLiNER2, achieving an overall F1-score of 0.488 against a manually annotated gold standard. For RE, closed-form extraction is performed by Claude, achieving an overall F1-score of 0.961. The resulting graph encodes anatomical entities and body movement terms as nodes, with edges derived from experimental conclusions in the source abstracts, with the goal of supporting hypothesis generation about biomechanical cause-and-effect chains.

1 Introduction

A recurring insight in biomechanics research is that the location of pain is often not the location of the underlying problem. Spinal pain may originate from hip imbalance; knee pain may trace back to abnormal foot posture. These proximal-cause relationships constitute what is commonly referred to as the "kinetic chain," and investigating them requires experiments that are costly in both time and resources.

This project addresses the question of whether a knowledge graph constructed from existing literature could help researchers identify plausible hypotheses prior to committing to an experiment. The approach is inspired by biomedical ontology construction (Arsenyan et al., 2024), but is particularly well-suited to biomechanics, where anatomical structures are naturally interconnected and their relationships are grounded in measurable kinematics.

As a proof-of-concept, a small knowledge graph is constructed from 19 abstracts drawn from biomechanics studies. Anatomical entities and body movements are encoded as nodes; edges represent experimental conclusions extracted from the abstracts. Graph traversal enables a researcher to discover indirect connections across studies — for example, that movement A is associated with symptom B in one paper, while symptom B contributes to condition C in another.

The entire pipeline is zero-shot: no model is fine-tuned and no labeled training data is produced. GLiNER2 (Zaratiana et al., 2025) and Claude are used as-is, which keeps the implementation lightweight but bounds performance to what general-purpose models can infer from domain-specific text.

This report is structured as follows. Section 2 introduces the theoretical background. Section 3 describes the dataset. Section 4 describes the pipeline. Section 5 presents evaluation results and graph analysis. Sections 6 and 7 discuss limitations and conclusions.

2 Theory

The pipeline is built around two core NLP tasks: Named Entity Recognition (NER) and Relation Extraction (RE).

2.1 Zero-Shot Named Entity Recognition

The primary design distinction from standard NER is the extraction schema. Rather than adopting a general-purpose taxonomy, the entity types are defined with respect to what is informative in an experimental biomechanics abstract. The five-type schema (Anatomical Entity, Symptom, Terms of Body Movements, Population, Measurement) is detailed in Appendix E. Beyond extraction, normalization collapses surface variants to canonical forms, and entity specification assigns each movement or symptom to its corresponding anatomical structure.

2.2 Zero-Shot Relation Extraction

RE adopts a *closed-form* schema in which the set of permissible relation types is fixed in advance. This contrasts with open-form RE, which tends to produce inconsistent free-text labels that cannot be directly mapped to graph edges. The five relation types and their rationale are provided in Appendix E.

2.3 Evaluation

All stages are evaluated against a manually annotated gold standard using precision, recall, and F1-score. For NER, matching is performed at the form level: each entity is counted once per abstract and field, regardless of the number of occurrences in the text. For RE, exact triple match is applied: a predicted triple is credited only if the subject, relation type, and object all correspond to the gold standard.

3 Data

The dataset comprises 19 biomechanics studies selected manually, from which abstracts were extracted as the primary pipeline input. Full paper text was excluded, as abstracts tend to state conclusions directly, making relation extraction more tractable.

The selected studies predominantly examine associations between symptoms and their underlying biomechanical causes, validated through controlled experiments comparing two population groups (e.g., patients vs. healthy controls). This structure is well-suited to `contributes_to` and `associated_with` extraction.

Citation count and publication year were deliberately excluded as selection criteria. Novel hypotheses may emerge from less prominent or older studies; restricting to high-citation or recent work could unnecessarily limit the scope of hypothesis generation.

4 Method

4.1 Models

The primary extraction model is **GLiNER2** (`fastino/gliner2-large-v1`), a DeBERTa-v3-large-based information extraction model with approximately 340M parameters, supporting zero-shot extraction against arbitrary label schemas.

For entity specification, three domain-specific BERT-based encoders were evaluated: **BioBERT** (`dmis-lab/biobert-base-cased-v1.1`), **PubMedBERT** (Gu et al., 2020) (`microsoft/BiomedNLP-BiomedBERT-base-uncased-abstract-fulltext`), and **SciBERT** (`allenai/scibert_scivocab_uncased`) (Beltagy et al., 2019), alongside a bag-of-words hashing baseline. **GPT-5** was used for NER pre-annotation and **Claude** (`claude-opus-4-6`) for RE.

4.2 Named Entity Recognition

General Extraction. GLiNER2 is applied to all 19 abstracts using the five-type schema (Appendix E), returning a list of labelled entity spans per abstract.

Normalization. Extracted entities are normalized to canonical forms via a local dictionary, covering abbreviation expansion, lemmatization of irregular forms, and disambiguation of surface variants that refer to the same concept.

Gold Standard Creation. The NER gold standard was constructed semi-manually using a custom HTML annotation tool (Appendix C). GPT-5 pre-annotations were loaded as a starting point; a human annotator then reviewed, corrected, and supplemented each abstract. The final export serves as the gold standard for all NER evaluations.

4.3 Entity Specification

Each *Symptom* and *Terms of Body Movements* entity is assigned to an anatomical structure to ground the graph's anatomical link edges. GLiNER2 is applied in a second round using question-style labels (Appendix E), though its architecture does not process question semantics as a generative model would, limiting effectiveness. As an alternative, the task is framed as nearest-neighbour search: entity forms and candidate anatomy labels are encoded by a BERT-based model, and the most similar candidate is selected by cosine similarity.

Population entities are additionally classified into sub-groups by age, sex, activity level, and clinical status, based on the assumption that experimental results vary across subject groups.

4.4 Relation Extraction

Open-form RE produces free-text descriptions unsuitable for graph construction, while existing biomedical RE models carry domain-specific relation schemas that do not transfer cleanly to biomechanics. Closed-form RE with Claude avoids both issues: the model receives the five relation types

explicitly and returns structured JSON triples. The full prompt is in Appendix D.

Claude receives the abstract text together with the normalized entity list and is constrained to use only those entity forms as triple endpoints. RE gold standard creation followed the same semi-manual process as NER, using a dedicated annotation tool (Appendix C).

4.5 Graph Construction

The graph is constructed from normalized entities, anatomical co-occurrence data, and RE triples. Three node types are used: Anatomical Entity (blue hexagon), Symptom (yellow diamond), and Movement (red ellipse). Two edge types are rendered: `anatomical_link` (green dashed), connecting anatomy nodes to co-occurring symptom or movement nodes; and `contributes_to` / `associated_with` (yellow/blue), derived from RE triples. Remaining relation types are stored in the graph data but not visualized, as they would add clutter without aiding hypothesis traversal.

Two strategies were compared for anatomical link construction: (1) co-occurrence enrichment, yielding 26 edges, and (2) `located_in` RE triples, yielding 8 edges. Each logical edge also stores supporting metadata: source abstract, evidence snippet, and associated population entities.

5 Results

5.1 NER Evaluation

Table 1: Round 1 NER performance of GLiNER2 against the manual gold standard. Matching is at the form level (one count per entity per abstract and field).

Entity Type	P	R	F1
Anatomical Entity	0.667	0.534	0.593
Measurement	0.473	0.582	0.522
Population	0.385	0.170	0.236
Symptom	0.694	0.486	0.571
Body Movements	0.647	0.349	0.454
Overall	0.576	0.424	0.488

The overall F1 of 0.488 masks notable variation across entity types. *Population* exhibits the lowest performance (F1 = 0.236): the model consistently fails to identify long-span, contextually implicit descriptions such as age-delimited cohort definitions, which fall outside the distribution of entities GLiNER2 was pre-trained to recognize (see Appendix A). Visual inspection of the NER output further confirms that span detection is generally reasonable, but type assignment is unreliable under this domain-specific schema.

For *Measurement*, predicted count (112) exceeds gold (91), suggesting over-extraction. This likely reflects annotation ambiguity at the boundary between measurement instruments and statistical evidence values, which were not annotated consistently during gold standard construction.

The gold standard itself carries a further source of noise: the boundary between *Symptom* and *Terms of Body Movements* is genuinely ambiguous for a subset of entities, and without inter-annotator agreement checks, labelling decisions were not always consistent.

5.2 Entity Specification Evaluation

Table 2: Entity specification evaluation: anatomical location assignment for *Symptom* and *Terms of Body Movements* entities (70 cases from `golden_standard.csv`).

Method	P	R	F1
GLiNER2 Round 2	0.563	0.129	0.209
Hashing Baseline	0.266	0.243	0.254
BioBERT	0.032	0.014	0.020
PubMedBERT	0.032	0.014	0.020
SciBERT	0.065	0.029	0.040

All methods perform poorly. The bag-of-words hashing baseline achieves the highest recall (24.3%), outperforming all three domain-specific BERT encoders. The BERT models exhibit reasonable coverage but near-zero precision, producing confident predictions that are predominantly incorrect.

The most likely explanation is a vocabulary mismatch. The gold standard employs high-level anatomical labels (*Foot*, *Knee*, *Spine*), whereas the candidate pool consists of raw normalized entity forms from the abstracts (e.g., *medial longitudinal arch*, *tibialis anterior*). A BERT encoder representing "anterior knee pain" assigns higher cosine similarity to "tibialis anterior" than to "knee," an outcome that is semantically defensible but inconsistent with the gold standard. The hashing baseline, relying on surface token overlap, more reliably recovers the correct coarse label.

GLiNER2 exhibits the inverse pattern: low coverage but reasonable precision when a prediction is made, indicating that the model abstains on the majority of cases.

5.3 Relation Extraction Evaluation

Table 3: RE performance of Claude (`claude-opus-4-6`) against the manual gold standard (exact triple match). 163 gold relations across 19 abstracts.

Relation Type	P	R	F1
`associated_with`	0.931	0.985	0.957
`contributes_to`	0.963	1.000	0.981
`located_in`	0.853	1.000	0.921
`measured_by`	0.968	1.000	0.984
`prevalence_in`	1.000	1.000	1.000
Overall	0.931	0.994	0.961

RE performance is strong (F1 = 0.961), in sharp contrast to NER. The only underperforming type is `located_in` (P = 0.853), where five false positives arise from Claude inferring anatomical location from contextual cues rather than explicit textual statements, which the gold standard does not credit. `prevalence_in` achieves perfect scores, reflecting the structural unambiguity of the pattern "condition X was observed in population Y."

These results should be interpreted with the caveat that Claude receives the normalized entity list as input and is constrained to use those forms as triple endpoints. Strong performance therefore indicates that, *given correctly identified entities*, the relation schema is well-suited to the domain. The primary bottleneck in end-to-end pipeline quality remains the NER stage.

5.4 Graph Analysis

The final graph contains **198 nodes** and **159 edges**: 83 Anatomical Entity nodes, 54 Symptom nodes, 61 Movement nodes, approximately 100 `anatomical_link` edges, and 59 `contributes_to` / `associated_with` edges. Visual evidence for the problems described below is provided in Appendix B.

Island problem. The majority of nodes are connected only within a single abstract, with no paths traversing more than one study. Cross-abstract connectivity is a minimum requirement for hypothesis generation, and 19 abstracts is insufficient to produce it at the concept level.

NER quality propagation. Errors in the NER stage propagate directly to the graph. Compound entities such as *femoral rotation* are frequently misclassified as Anatomical Entity rather than Terms of Body Movements, causing them to appear as anatomy nodes and excluding them from `contributes_to` and `associated_with` edges.

Normalization gaps. Semantically equivalent entities persist as distinct nodes when the normalization dictionary lacks the corresponding rule — for example, *quadricep* and *leg muscle*, or multiple nodes representing variants of flat foot. In the absence of an external reference vocabulary, normalization coverage is limited to manually curated cases. breaks right before a citation number or parenthe

Uninformative anatomical links. Some `anatomical_link` edges are structurally valid but semantically uninformative: entities co-occurring in the same abstract are connected regardless of whether the link is meaningful for hypothesis generation.

What works. Despite these limitations, the graph contains well-connected regional subgraphs in which symptoms and movements are linked across abstracts via RE-derived edges. Each such edge stores supporting metadata including the source abstract, an evidence snippet, and population context tags, enabling a reader to assess the provenance and applicability of a given connection.

Strategy comparison. RE-based anatomical links (strategy 2, 8 edges) are more precise than co-occurrence links (strategy 1, 26 edges), as each is grounded in explicit textual evidence. Relying on RE for all edge types would produce a cleaner graph at the cost of requiring greater RE coverage.

6 Discussion

This project achieves strong RE quality but falls short on NER and data volume, which are the two requirements a hypothesis-generation graph most depends on.

Excluding LLMs from the NER stage to reduce cost proved to be a false economy: the engineering effort invested in annotation tools, comparison experiments, and iterative debugging of GLiNER2

likely exceeded the cost of a GPT-5-based NER approach, while still yielding below-median F1.

The entity specification experiment demonstrates that domain-specific BERT encoders do not generalize well when the candidate vocabulary is uncontrolled. Surface token overlap outperformed contextual embeddings, suggesting that a structured anatomy ontology as the candidate pool would be a more principled starting point for future work.

7 Conclusion

This work presents a zero-shot pipeline for biomechanical knowledge graph construction, encompassing NER, normalization, entity specification, relation extraction, and graph construction. RE performs well; NER is the limiting factor. The resulting graph is too sparse to support systematic cross-study hypothesis traversal, primarily due to data volume and NER quality.

The underlying task — accelerating hypothesis generation in biomechanics via knowledge graphs — remains a compelling research direction. The zero-shot framing eliminates the need for domain-specific training data, but does not eliminate the need for manual effort: annotation tools, evaluation infrastructure, and normalization resources all require substantial investment. A larger corpus and a controlled entity vocabulary represent the most impactful directions for future work.

References

Vahan Arsenyan, Spartak Bughdaryan, Fadi Shaya, Kent Wilson Small, and Davit Shahnazaryan. 2024. Large language models for biomedical knowledge graph construction: information extraction from emr notes. In *Proceedings of the 23rd Workshop on Biomedical Natural Language Processing*, pages 295–317.

Iz Beltagy, Kyle Lo, and Arman Cohan. 2019. Scibert: A pretrained language model for scientific text. In *EMNLP*. Association for Computational Linguistics.

Yu Gu, Robert Tinn, Hao Cheng, Michael Lucas, Naoto Usuyama, Xiaodong Liu, Tristan Naumann, Jianfeng Gao, and Hoifung Poon. 2020. Domain-specific language model pretraining for biomedical natural language processing.

Urchade Zaratiana, Gil Pasternak, Oliver Boyd, George Hurn-Maloney, and Ash Lewis. 2025. Gliner2: An efficient multi-task information extraction system with schema-driven interface. *Preprint*, arXiv:2507.18546.

A NER Result Visualization

The NER evaluation includes a custom HTML visualization that highlights predicted and gold entities side by side in the abstract text. Gold-only and predicted-only spans are rendered in distinct colours to make field-level mismatches immediately visible.

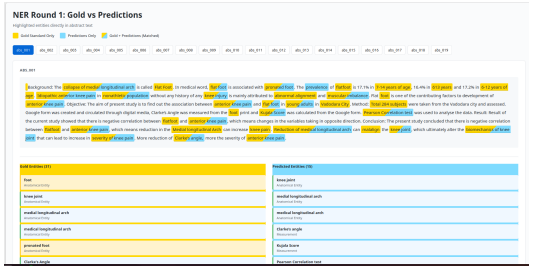
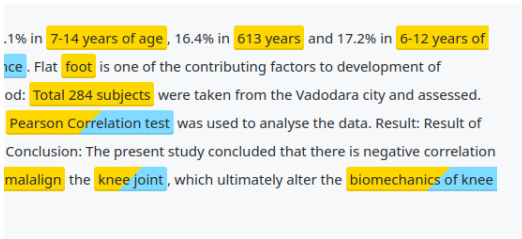


Figure 1: NER visualization: abstract text with colour-coded entity highlights. Gold-standard and predicted spans are overlaid; colour indicates the assigned field.

The right panel displays normalized entity lists for both gold and predicted, grouped by field, facilitating identification of missed versus mis-labelled entities.

Figure 2: Side-by-side normalized entity comparison between gold standard and GLiNER2 predictions, grouped by entity type.



breaks right before a citation number or parenthe

Figure 3: Example of a Population entity missed by GLiNER2. Long-span, contextually defined descriptions are consistently under-detected.

B Graph Visualization

The graph viewer is an interactive HTML tool built with vis.js. Node shapes encode type: blue hexagon (Anatomical Entity), yellow diamond (Symptom), red ellipse (Movement). Edge colours encode relation type: green dashed (`anatomical_link`), yellow solid (`contributes_to`), blue solid (`associated_with`).

B.1 Graph Overview and Edge Information

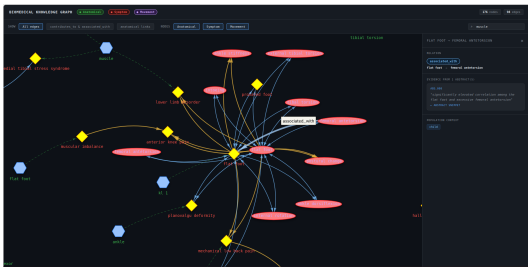


Figure 4: Graph with the *flat foot* → *femoral antetorsion* edge selected (`associated_with`, abs_008, population: *child*). The central cluster illustrates a well-connected subgraph; isolated peripheral nodes exemplify the island problem.

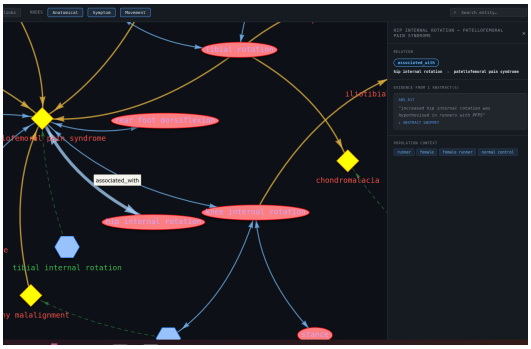


Figure 5: *Hip internal rotation* → *patellofemoral pain syndrome* (`associated_with`, abs_017). Population tags (*runner*, *female*, *female runner*, *normal control*) and an evidence snippet are shown in the right panel.

B.2 Graph Problems

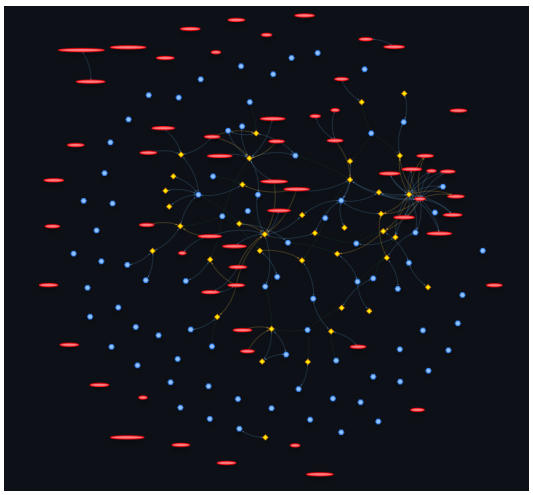


Figure 6: Full graph view. The majority of nodes are isolated, with meaningful connections concentrated in a small central cluster, confirming that cross-abstract paths are absent.

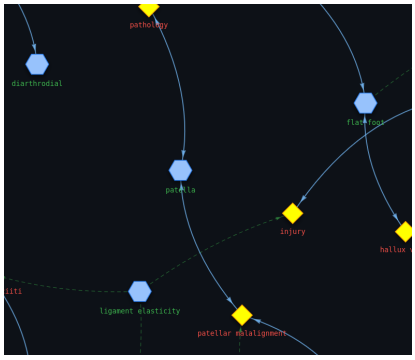


Figure 7: Misclassification: *ligament elasticity* and *flat foot* are rendered as Anatomical Entity nodes (blue hexagons) rather than Symptom or Movement nodes, excluding them from relation edges.

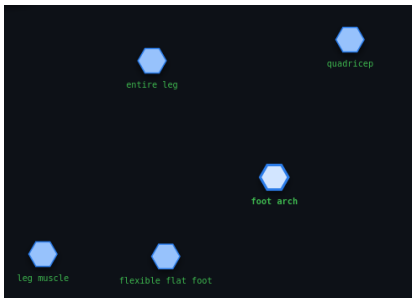


Figure 8: Normalization gaps: *entire leg*, *leg muscle*, and *quadricep* persist as three separate anatomy nodes. *Flexible flat foot* and *foot arch* exist alongside the main *flat foot* node.

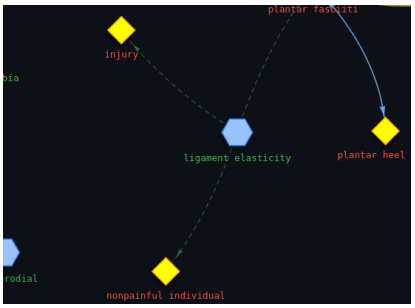


Figure 9: Uninformative anatomical links: *ligament elasticity* is connected to *injury*, *plantar heel*, and *nonpainful individual* via co-occurrence, without semantic relevance to hypothesis generation.

C Annotation Tools

C.1 NER Annotation Tool

The NER annotation tool is a self-contained HTML file generated by `src/tools/generate_annotation_tool.py`. GPT-5 pre-annotations are loaded as a starting point. The interface presents a category selector with colour-coded buttons and per-type entity counts, the abstract text in the centre, and a running annotation list on the right. Spans are marked by selecting a category and dragging across tokens; an undo button removes the last annotation. Completed annotations are exported as JSON.

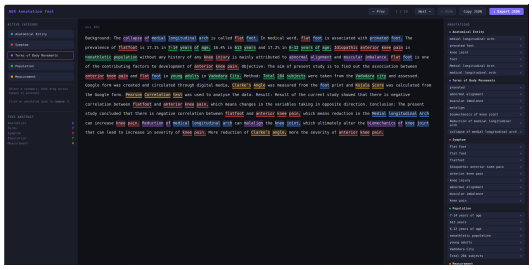


Figure 10: NER annotation tool. Left: entity type selector with per-type counts. Centre: abstract text with highlighted spans. Right: annotation list for the current abstract.

C.2 RE Annotation Tool

The RE annotation tool is generated by `src/tools/generate_relation_annotation_tool` Claude's extracted triples are pre-loaded for review; the annotator accepts, rejects, or edits each triple. New triples are constructed via entity dropdowns populated from the recognized entity forms of the current abstract, enforcing the constraint that all triple endpoints correspond to known entities.

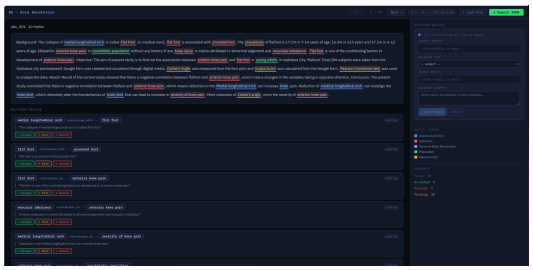


Figure 11: RE annotation tool. Pre-loaded Claude extractions are listed for review. The right panel supports construction of new triples via entity dropdowns and evidence text editing.

D Prompts

D.1 NER Prompt (GPT-5 Gold Standard)

Listing 1: NER extraction prompt used with GPT-5.

```
You are an expert in biomechanics and
    sports rehabilitation.
Your task is to extract all
    biomechanical and clinically
    relevant
entities from a given abstract and
    classify them into the following
categories.

CATEGORY DEFINITIONS:
1. Anatomical Entity - Specific body
    structures, regions, or systems.
   Examples: medial longitudinal arch,
       knee joint, tibia
2. Symptom / Condition - Clinical
    conditions or biomechanical
    abnormalities.
   Examples: anterior knee pain, flat
       foot, plantar fasciitis
3. Terms of Body Movements - Kinematic
    or kinetic descriptors of motion.
   Examples: pronation, dorsiflexion,
       valgus
4. Population - Participant groups by
    age, activity level, or status.
   Examples: young adults, nonathletic
       population, elite athletes
5. Measurement - Tools, methods, or
    statistical values used to draw
    conclusions.
   Examples: Clarke's Angle, Meary's
       angle, p-value

RULES:
- Extract entities verbatim as they
    appear in the text.
- If a category has no entities,
    return an empty list.
- Do not infer entities not present in
    the text.
- Process each abstract independently.

OUTPUT FORMAT (strict JSON, no
    markdown):
{
  "Abstract 1": {
    "entities": {
```

```
      "Anatomical Entity": [],
      "Symptom": [],
      "Terms of Body Movements": [],
      "Population": [],
      "Measurement": []
    }
  }
}
```

D.2 RE Prompt (Claude)

Listing 2: RE extraction prompt used with Claude.

```
Role: You are a Knowledge Graph
    Specialist specializing in
    Biomedical
Relation Extraction. Identify semantic
    relationships between predefined
entities in clinical abstracts.

Task: Extract relation triples
    (Subject, Relation, Object) using
    ONLY
the entity forms provided in the input.

Candidate Relations:
1. associated_with - A general
    clinical or anatomical link.
2. contributes_to  - Entity A is a
    risk factor or cause for entity B.
3. measured_by     - A symptom or
    feature is evaluated using a tool.
4. prevalence_in   - A condition
    occurs within a specific
    population.
5. located_in      - Anatomical
    localization.

Rules:
- Use ONLY the entity_form values from
    the input as subject and object.
- If no relationship is explicitly
    stated, DO NOT extract it.
- Base extraction strictly on the
    provided text.

Output Format (JSON array):
[
  {
    "subject": "Entity A",
    "relation": "RELATION_TYPE",
    "object": "Entity B",
    "evidence": "Snippet from abstract
        text"
  }
]
```

E Schemas

E.1 NER Entity Schema

Table 4 lists the five entity types used by both GLiNER2 and GPT-5, along with descriptions and examples.

Table 4: NER entity schema used by both GLiNER2 and GPT-5.

Entity Type	Description and Examples
Anatomical Entity	Any biological body structure: bones, joints, muscles, arches. *E.g., medial longitudinal arch, knee joint, tibia.*
Symptom	Clinical conditions, pain presentations, or biomechanical abnormalities. *E.g., anterior knee pain, flat foot, plantar fasciitis.*
Terms of Body Movements	Kinematic or kinetic descriptors of motion, posture, or alignment. *E.g., ankle dorsiflexion, hip internal rotation, valgus.*
Population	Participant groups defined by age, activity level, or clinical status. *E.g., young adults, elite athletes, children aged 7–14.*
Measurement	Tools, methods, scores, or statistical values used to draw conclusions. *E.g., Clarke's Angle, Meary's angle,* $p < 0.05$.

E.2 Entity Specification Schema (Round 2)

Table 5 shows the question-style labels used in the second extraction round for Symptom, Movement, and Population entities.

Table 5: Round 2 specification schema for Symptom, Terms of Body Movements, and Population entities.

Field	Label	Question / Description
Symptom	anatomical structure	Where is the symptom located?
	symptom	What is the symptom or condition?
Terms of Body Movements	anatomical structure	Which structure is involved?
	movement	What is the movement or measure?
Population	age	Age, age range, or developmental stage.
	level	Athlete level or non-athlete.
	clinical status	Condition or case/-control status.
	sex	Sex or gender.
	study group	Named cohort or study arm.

E.3 RE Relation Schema

- `associated_with`: a general clinical or experimental link.
- `contributes_to`: entity A is a risk factor for entity B.
- `located_in`: anatomical localization.
- `measured_by`: a symptom or feature is quantified by a tool.
- `prevalence_in`: a condition occurs within a specific population.

The first two types capture experimental findings; `located_in` and `measured_by` encode supporting anatomical and methodological context; `prevalence_in` links conditions to the population under study.

F Use of Large Language Models

Large language models were used at several stages of this project. During the initial scoping phase, an LLM was consulted to identify relevant studies and narrow the topic to musculoskeletal biomechanics. In the implementation phase, LLM assistance was used to manipulate and transform JSON data structures, and to reorganise exploratory Jupyter

notebook code into documented, modular Python scripts. Custom HTML tools for NER and RE annotation, as well as the interactive graph visualisation, were built with LLM assistance. In the final reporting phase, an LLM was used to verify LaTeX formatting and to improve the clarity and academic register of the written text.

All pipeline design decisions, gold standard annotations, result interpretations, and analytical conclusions are the author's own.